Lyapunov Perception Contracts for Operating Design Domain

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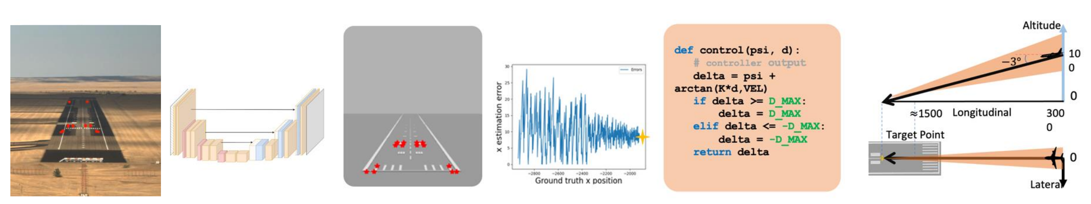
INTRODUCTION

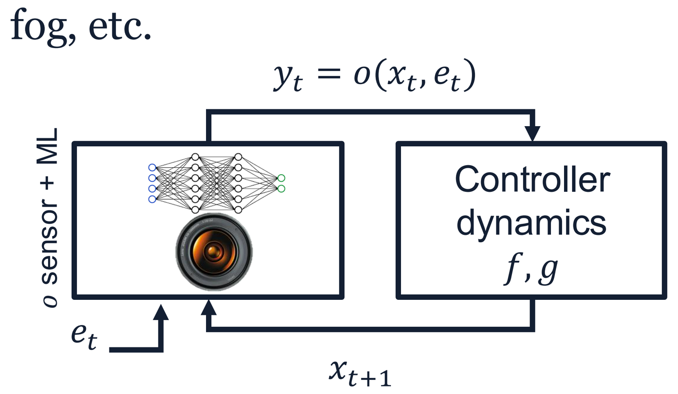
Problem: Reliable Visual Autonomy
We study the problem of providing high level of **correctness assurances** autonomous systems that use machine learning models (e.g., CNNs, VITs, etc.) for state estimation or for pixels-to-torque control.

An Example: Visual Auto-landing
Pipeline uses cameras generating *high-dimensional* signals (images) which are processed by algorithms (CNNs, PnP), to output a *low-dimensional state estimates* (e.g., airplane pose) which drive the controller. **Correctness specified** as state invariants, barrier certificates, etc.



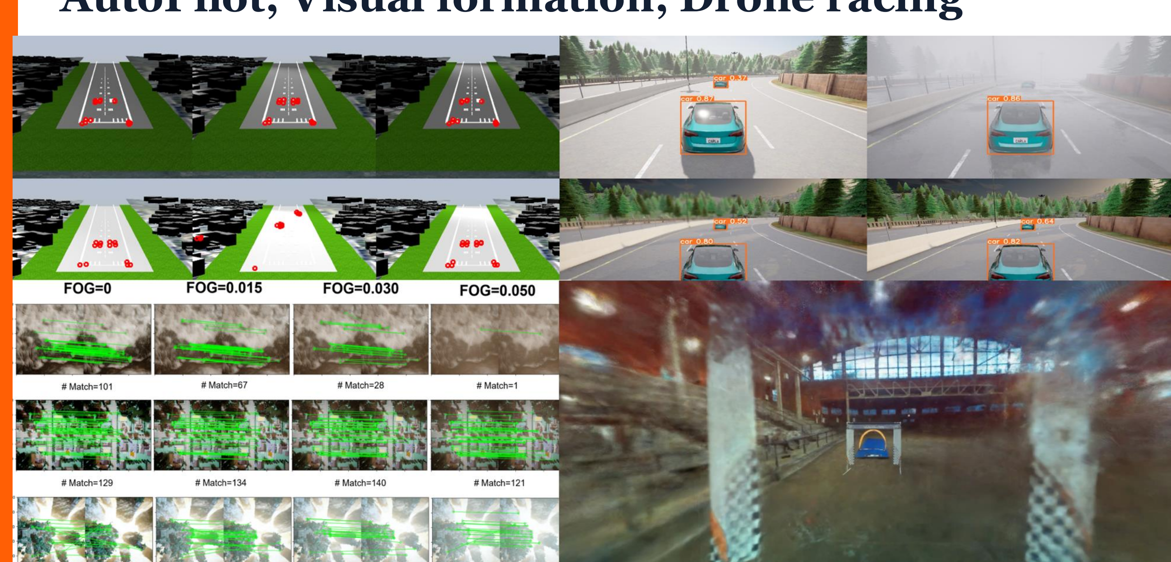
Camera UNet Keypoints PnP Control Safety envelop

Abstract Closed-loop System: A coarse observer $h(x_t, \varepsilon_t)$ with a parameterized environment $e_t \in E$, e.g., lighting, fog, etc.



Examples and Assumptions

AutoPilot, Visual formation, Drone racing



Finitely Parameterized Environment

Perception dropped-in for state estimation, i.e., with ground truth state $o(x_t, e_t) = x_t$ we have a correctness proof of the system

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Problem Statement

Consider a vision–based system $\dot{x} = f(x, g(o(x,e)))$, where model dynamics and controller f,g are white-box, and perception pipeline h is black-box.

Given a candidate Lyapunov function V and a target set X^*, we are interested in finding a subset $E_0 \subseteq E$ of the environment (ODD), in which the system is asymptotical stable w.r.t the set X^*.

Asymptotical Stability of Set-Valued System

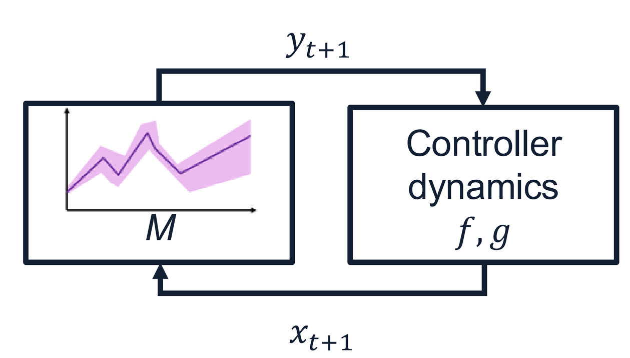
Consider a system $\dot{x} = F(x)$, where F is a set-valued mapping $F: X \to 2^X$. The following theorem provides a sufficient condition for asymptotical stable of above system.

Theorem. : For system $\dot{x} = F(x)$, given
1) a nonempty, compact set $X^* \subseteq X$;
2) a continuous, differentiable function $V: X \to \mathbb{R}_{\geq 0}$ such that $V(x) > c$ for all $x \in X^*/X$, and $V(x) \leq c$ for all $x \in X^*$;
3) a continuous function $W: X \to \mathbb{R}_{\geq 0}$ such that $W(x) = c$ iff $x \in X^*$.

If $\forall x \in X$, $f(x) \in F(x)$, it satisfies
$$\nabla V(x) \cdot f(x) \leq -W(x)$$
Then X^* is asymptotical stable and V is a Lyapunov function for the system.

Lyapunov Perception Contracts [1,2,3]

We handle the black-box perception pipeline by computing its over-approximation. If the over-approximated system is stable, the stability of original system follow.



This is an abstraction of the original closed-loop system that is sufficient for proving asymptotical stable.

Definition. For system $\dot{x} = f(x, g(o(x,e)))$, given a target set X^*, candidate Lyapunov function V and an over-approximation of perception $M: X \to 2^Y$ is called E_0, X^*-Lyapunov Perception Contract if
1. **Conformance**. $h(x, \varepsilon) \in M(x), \forall \varepsilon \in E_0, x \in X$
2. **Correctness**. V is Lyapunov function w.r.t set X^* for system $\dot{x} = f(x, g(M(x)))$

Proposition. If there exists a Lyapunov perception contract M for h, X^*, E_0 then original system is asymptotical stable w.r.t set X^*.

Learning Operating Design Domains

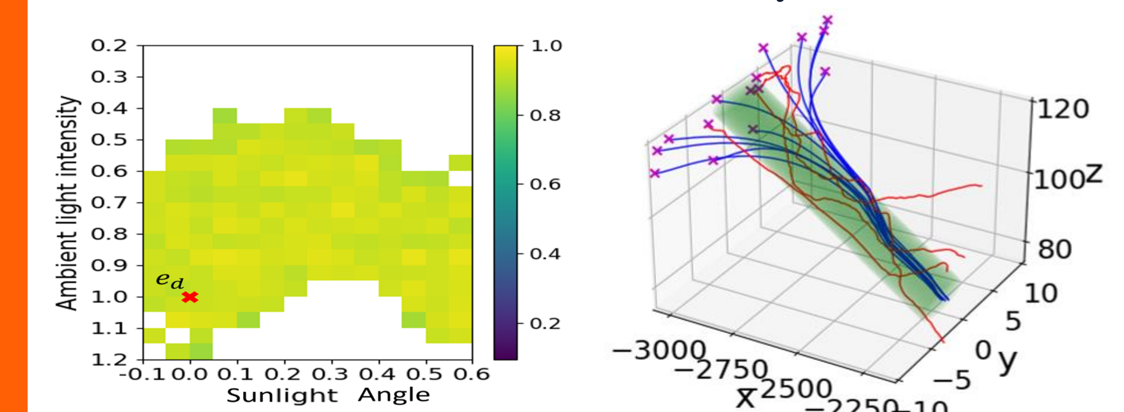
Function LearnCtrt $(\tilde{X}, \tilde{E}, o, pr, \epsilon, \delta)$:
$n \leftarrow -\frac{\ln \delta}{2\epsilon^2}$
$\{\langle x_i, e_i \rangle\}_{i=1}^n \leftarrow \text{Sample}_{\mathcal{D}}(\tilde{X}, \tilde{E}, n)$
$\hat{y}_1, \ldots, \hat{y}_n \leftarrow o(x_1, e_1), \ldots, o(x_n, e_n)$
$L_c \leftarrow \{\langle x_1, \hat{y}_1 \rangle, \ldots, \langle x_n, \hat{y}_n \rangle\}$
$M_c \leftarrow \text{Regression}(L_c)$
$L_r \leftarrow \{\langle x_i, |\hat{y}_1 - M_c(x_i)| \rangle\}_{i=1}^n$
$M_r \leftarrow \text{QuantileRegression}(L_r, pr+\epsilon)$
return $\langle M_c, M_r \rangle$

LearnCtrt aims to train a mapping that outputs a spherical over-approximation of actual perception output with a specified probability guarantee.

It samples (x_i, e_i) from state and environment domain, and renders perception output $\hat{y}_i$.

Then it computes the mapping to over-approximation center M_c, over-approximation radius M_r by regression under given probabilistic guarantee.

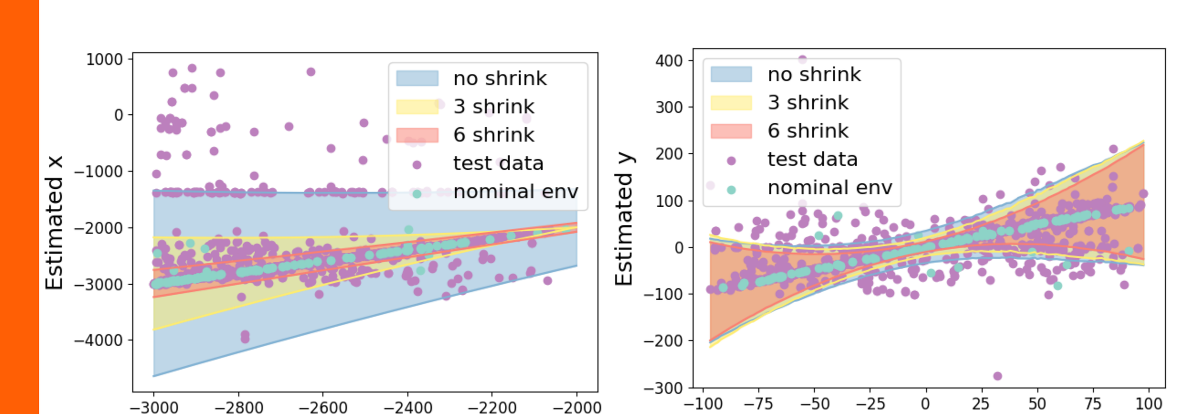
Figures below shows computed ODDs for auto-landing and reachable states of the abstract system.



Function DaReLPC $(\tilde{X}, E, e_d, X^*, o)$:
Parameters: pr, ϵ, δ, V
$\tilde{E} \leftarrow E$
$M_{\tilde{E}} \leftarrow \text{LearnCtrt}(\tilde{X}, \tilde{E}, o, pr, \epsilon, \delta)$
while *True* **do**
  if $\text{CheckLyap}(\tilde{X}, M_{\tilde{E}}, V, X^*)$ **then**
    break
  $\tilde{E} \leftarrow \text{ShrinkEnv}(\tilde{E}, e_d)$
  $M_{\tilde{E}} \leftarrow \text{LearnCtrt}(\tilde{X}, \tilde{E}, o, pr, \epsilon, \delta)$
return $\langle M_{\tilde{E}}, \tilde{E} \rangle$

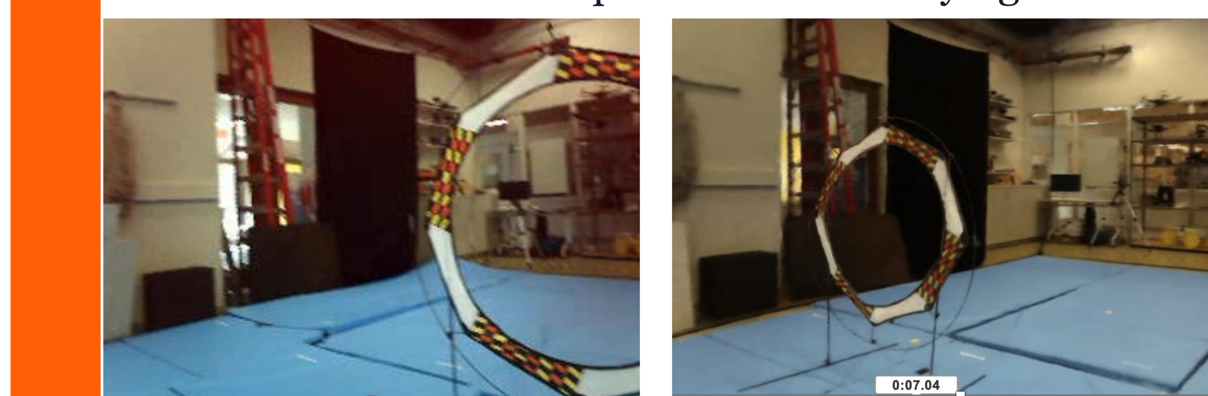
DaReLPC aims to find the perception contract $M_{\tilde{E}}$ satisfying certain Lyapunov condition by shrinking ODD ($\tilde{E}$).

Figures below visualizes the learned perception contract of different number of shrinking on two dimensions.

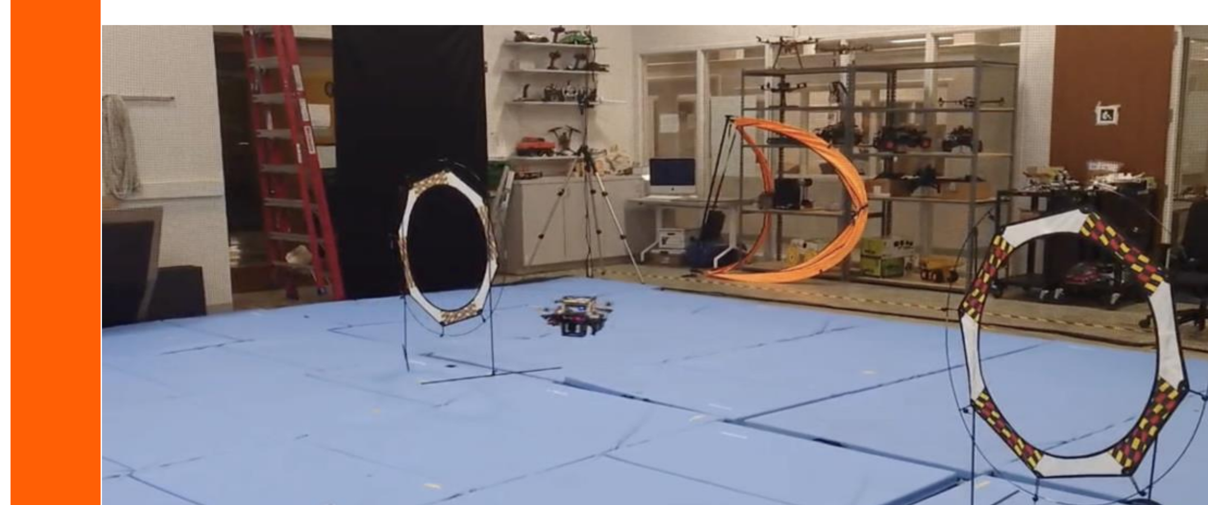


Sim2Real Visual Policies from Neural Reconstructions: A Preview

Neural scene representations can be used to train visual control policies. We have been able to demonstrate this Sim2Real transfer with quadrotors in our flying arena



Actual scene NeRF reconstruction



CONCLUSIONS

Combining formal (symbolic) & statistics we can provide assurances and explanations for visual autonomy under set-valued Lyapunov theorem.

Lyapunov Perception contracts provide *sufficient* asymptotical stability condition on ML-based state estimators for system-level correctness

References

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